

SIOUX GATEWAY, IA, USA

WMO: 725570

Lat: 42.400N

Lon: 96.383W

Elev: 336

StdP: 97.35

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14943

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.7	-19.1	-25.4	0.4	-21.1	-23.2	0.5	-18.7	16.2	-7.4	14.5	-8.9	4.0	320	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	33.8	24.1	32.3	23.6	30.8	22.9	26.1	31.5	25.2	30.4	24.2	29.2	6.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	20.4	29.8	23.5	19.1	28.8	22.6	18.0	27.8	83.1	31.6	78.8	30.5	74.8	29.2	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.2	9.8	-26.3	36.6	2.6	1.9	-28.1	37.9	-29.7	39.0	-31.1	40.1	-33.0	41.5		
(5)			-26.5	28.0	2.6	0.7	-28.3	28.5	-29.8	28.9	-31.3	29.3	-33.1	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-6.5	-4.1	2.9	9.8	16.2	21.8	23.9	22.4	18.2	10.7	2.9	-4.1
	DBStd	11.94	6.84	6.96	6.89	5.53	4.91	3.77	3.19	3.04	4.64	5.25	5.76	6.28
	HDD10.0	1934	511	395	237	69	6	0	0	0	2	54	222	438
	HDD18.3	3716	769	628	479	261	101	12	2	4	59	241	463	697
	CDD10.0	1782	0	1	17	63	199	354	430	384	248	78	9	0
	CDD18.3	520	0	0	1	4	36	116	173	130	54	6	0	0
	CDH23.3	5392	0	0	15	97	430	1216	1791	1168	587	86	3	0
	CDH26.7	2003	0	0	2	28	153	462	732	403	199	23	0	0
Wind	WSAvg	4.5	4.9	5.0	5.1	5.5	5.0	4.3	3.5	3.3	3.9	4.4	4.7	4.7
Precipitation	PrecAvg	700	17	19	46	70	95	104	83	90	72	52	29	22
	PrecMax	1053	69	68	150	171	190	423	262	257	246	174	116	73
	PrecMin	364	3	2	1	12	22	14	9	3	8	2	0	3
	PrecStd	142	13	14	33	40	39	64	55	57	49	40	25	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	16.4	25.4	30.2	33.8	35.1	36.3	34.8	33.9	30.1	22.5	13.3
		MCWB	7.0	10.6	14.3	17.4	20.5	23.9	25.7	25.0	22.4	19.5	14.0	8.7
	2%	DB	8.1	11.6	21.1	26.1	30.4	33.0	33.9	32.3	31.1	25.6	18.4	9.7
		MCWB	4.1	7.3	13.2	15.5	19.5	23.0	25.4	24.8	21.6	16.8	11.9	5.7
	5%	DB	4.9	7.9	16.8	22.8	28.0	31.2	32.2	30.5	29.0	22.7	15.0	6.6
		MCWB	2.2	4.7	11.6	14.0	18.2	22.2	24.8	23.8	21.0	15.5	9.6	3.6
	10%	DB	2.5	5.0	13.5	19.9	25.4	29.4	30.6	28.9	26.9	20.1	11.9	4.2
		MCWB	0.9	2.8	8.9	12.3	17.1	21.3	24.0	22.7	19.7	13.9	7.7	2.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	11.3	16.9	18.7	22.9	26.0	27.8	26.9	24.5	21.1	15.4	10.0
		MCDB	12.2	15.5	22.2	26.7	30.0	32.2	32.9	32.3	30.2	25.7	19.9	12.4
	2%	WB	4.4	7.4	14.3	16.7	21.2	24.8	26.7	25.6	23.3	18.6	12.7	5.8
		MCDB	7.5	11.3	18.6	24.0	27.6	30.8	31.7	30.8	28.9	23.1	17.9	9.1
	5%	WB	2.4	4.8	11.8	14.9	19.6	23.7	25.6	24.7	22.3	16.6	10.2	3.8
		MCDB	4.6	7.8	17.2	21.7	25.8	29.3	30.8	29.3	27.4	21.2	14.4	6.3
	10%	WB	0.9	2.8	8.9	13.0	18.1	22.6	24.7	23.6	21.0	14.6	7.8	2.2
		MCDB	2.4	4.8	13.7	18.6	23.9	27.8	29.8	27.8	25.4	19.3	11.7	4.1
Mean Daily Temperature Range	5% DB	MDBR	10.0	10.3	11.5	13.0	12.5	11.9	11.5	11.5	13.5	13.1	11.7	9.8
		MCDBR	14.1	14.1	18.0	18.7	16.2	13.8	12.9	13.0	15.6	17.4	17.0	14.0
		MCWBR	10.6	10.2	10.9	9.7	7.5	6.3	6.3	6.7	7.5	9.4	10.7	10.4
	5% WB	MCDBR	13.1	13.4	16.7	15.8	13.6	11.7	11.5	11.5	12.8	14.2	15.6	12.9
		MCWBR	10.1	9.9	10.4	9.0	7.2	6.4	6.4	6.4	6.9	8.7	10.4	9.8
Clear-Sky Solar Irradiance	taub		0.279	0.291	0.323	0.387	0.413	0.417	0.427	0.422	0.374	0.330	0.294	0.282
	taud		2.431	2.423	2.406	2.241	2.240	2.291	2.264	2.271	2.369	2.478	2.498	2.445
	Ebn at Noon		873	918	923	880	863	858	845	836	855	860	849	835
	Edh at Noon		75	90	104	132	136	130	132	127	107	85	71	69
All-Sky Solar Radiation	RadAvg		1.94	3.00	3.82	4.89	5.72	6.39	6.60	5.55	4.49	3.11	2.17	1.58
	RadStd		0.18	0.24	0.30	0.38	0.42	0.47	0.34	0.40	0.30	0.38	0.14	0.13

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page